

ELEC2600 Homework 4 Solutions

Question 1

Let the lifetime of each device have mean μ and standard deviation $\sigma = 2$. By the Central Limit Theorem,

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) = \mathcal{N}\left(\mu, \frac{4}{n}\right).$$

(1.1)

We want

$$P(|\bar{X}_n - \mu| \leq 0.05) \geq 0.96.$$

Standardizing,

$$P\left(|Z| \leq \frac{0.05}{2/\sqrt{n}}\right) \geq 0.96.$$

Since $P(|Z| \leq z) = 0.96$, we need $z = z_{0.98} \approx 2.0537$. Hence

$$\frac{0.05\sqrt{n}}{2} \geq 2.0537 \implies n \geq \left(\frac{2 \times 2.0537}{0.05}\right)^2 \approx 6748.62.$$

Therefore, the required sample size is

$$\boxed{n = 6749}.$$

(1.2)

Now $\mu = 8$ and $n = 2500$, so

$$\bar{X}_{2500} \approx \mathcal{N}\left(8, \frac{4}{2500}\right),$$

with standard deviation

$$\frac{2}{\sqrt{2500}} = \frac{2}{50} = 0.04.$$

Thus

$$P(|\bar{X}_{2500} - 8| \leq 0.05) = P\left(|Z| \leq \frac{0.05}{0.04}\right) = P(|Z| \leq 1.25).$$

So

$$P(|\bar{X}_{2500} - 8| \leq 0.05) = 2\Phi(1.25) - 1 \approx 0.7887.$$

Therefore,

$$P(|\bar{X}_{2500} - 8| \leq 0.05) \approx 0.7887.$$

Question 2

Given a Poisson process with rate $\lambda = 80$ customers/hour.

For time interval t hours,

$$N(t) \sim \text{Poisson}(\lambda t).$$

(2.1)

The first 9 minutes is $t = 9/60 = 0.15$ hours, so

$$\lambda t = 80(0.15) = 12.$$

Hence

$$P(N(9 \text{ min}) = 10) = e^{-12} \frac{12^{10}}{10!}.$$

Therefore,

$$P = e^{-12} \frac{12^{10}}{10!} \approx 0.10484.$$

(2.2)

The next 21 minutes is $21/60 = 0.35$ hours, so its mean is

$$80(0.35) = 28.$$

By independent increments,

$$P(N(9 \text{ min}) = 10 \text{ and next 21 min has 30}) = \left(e^{-12} \frac{12^{10}}{10!}\right) \left(e^{-28} \frac{28^{30}}{30!}\right).$$

Therefore,

$$P = \left(e^{-12} \frac{12^{10}}{10!}\right) \left(e^{-28} \frac{28^{30}}{30!}\right) \approx 0.00710.$$

(2.3)

The waiting time of the first customer T_1 is exponential with rate 80/hour. For 1 minute,

$$t = \frac{1}{60} \text{ hour.}$$

Thus

$$P(T_1 < 1 \text{ min}) = 1 - e^{-80/60}.$$

Therefore,

$$P = 1 - e^{-4/3} \approx 0.73640.$$

(2.4)

$$\begin{aligned} P(1 < T_1 < 10 \text{ min}) &= P(T_1 > 1 \text{ min}) - P(T_1 > 10 \text{ min}) \\ &= e^{-80/60} - e^{-800/60}. \end{aligned}$$

Therefore,

$$P = e^{-4/3} - e^{-40/3} \approx 0.26360.$$

(2.5)

The second arrival time T_2 satisfies

$$P(T_2 < t) = P(N(t) \geq 2).$$

For $t = 3 \text{ minutes} = 3/60 = 0.05 \text{ hour}$, the mean is

$$\lambda t = 80(0.05) = 4.$$

Hence

$$\begin{aligned} P(T_2 < 3 \text{ min}) &= 1 - P(N(3 \text{ min}) = 0) - P(N(3 \text{ min}) = 1) \\ &= 1 - e^{-4} - 4e^{-4} = 1 - 5e^{-4}. \end{aligned}$$

Therefore,

$$P = 1 - 5e^{-4} \approx 0.90842.$$

Question 3

Let $I_i = 1$ if the i th draw is a Jack, and $I_i = 0$ otherwise. Since there are 4 Jacks in 52 cards,

$$p = P(I_i = 1) = \frac{4}{52} = \frac{1}{13}.$$

Each win is \$5, so

$$S_n = 5 \sum_{i=1}^n I_i.$$

Thus $\sum_{i=1}^n I_i \sim \text{Binomial}(n, 1/13)$.

(3.1)

$$E[S_{13}] = 5E\left[\sum_{i=1}^{13} I_i\right] = 5 \cdot 13 \cdot \frac{1}{13} = 5.$$

Also,

$$\text{Var}(S_{13}) = 25 \text{Var}\left(\sum_{i=1}^{13} I_i\right) = 25 \cdot 13 \cdot \frac{1}{13} \left(1 - \frac{1}{13}\right) = 25 \cdot \frac{12}{13}.$$

So

$$\boxed{E[S_{13}] = 5, \quad \text{Var}(S_{13}) = \frac{300}{13}.$$

Now

$$S_{20} = S_{13} + (S_{20} - S_{13}),$$

and the increment $S_{20} - S_{13}$ is independent of S_{13} . Therefore,

$$\text{Cov}(S_{13}, S_{20}) = \text{Cov}(S_{13}, S_{13}) + \text{Cov}(S_{13}, S_{20} - S_{13}) = \text{Var}(S_{13}).$$

Hence

$$\boxed{\text{Cov}(S_{13}, S_{20}) = \frac{300}{13}.$$

(3.2)

Winning \$30 in 15 games means exactly 6 Jacks in 15 draws. Hence

$$\boxed{P(S_{15} = 30) = \binom{15}{6} \left(\frac{1}{13}\right)^6 \left(\frac{12}{13}\right)^9.}$$

(3.3)

$S_{20} = 80$ means 16 Jacks in 20 games, while $S_{15} = 30$ means 6 Jacks in the first 15 games. The last 5 games can contribute at most 5 more Jacks, so this is impossible. Therefore,

$$\boxed{P(S_{20} = 80 \mid S_{15} = 30) = 0.}$$

(3.4)

Since the event is impossible,

$$P(S_{20} = 80 \cap S_{15} = 30) = 0.$$

(3.5)

Starting with \$10 and paying \$2 per game, after 5 games the net amount is

$$S_5 = 10 - 2(5) + 5W = 5W,$$

where $W \sim \text{Binomial}(5, 1/13)$ is the number of Jacks in 5 games.

We need $S_5 > 0$, i.e. $W \geq 1$. Thus

$$P(S_5 > 0) = 1 - \left(\frac{12}{13}\right)^5.$$

(3.6)

Starting with \$12 and paying \$2 per game, after 20 games the net amount is

$$S_{20} = 12 - 2(20) + 5W = -28 + 5W,$$

where $W \sim \text{Binomial}(20, 1/13)$.

We need

$$-28 + 5W > 0 \implies W \geq 6.$$

Therefore,

$$P(S_{20} > 0) = \sum_{k=6}^{20} \binom{20}{k} \left(\frac{1}{13}\right)^k \left(\frac{12}{13}\right)^{20-k}.$$

Question 4

For $i \geq 2$,

$$Y_i = X_i - \frac{1}{2}X_{i-1}.$$

(4.1) X_i i.i.d., $E[X_i] = 400$, $\text{Var}(X_i) = 8000$

Mean

$$E[Y_i] = E[X_i] - \frac{1}{2}E[X_{i-1}] = 400 - \frac{1}{2}(400) = 200.$$

So

$$\mu_Y = 200.$$

Autocovariance

Let $m = j - k$. Since the X_i are i.i.d.,

$$C_Y(m) = \text{Cov}(Y_j, Y_k) = \text{Cov}\left(X_j - \frac{1}{2}X_{j-1}, X_k - \frac{1}{2}X_{k-1}\right).$$

Only equal-time terms survive.

For $m = 0$,

$$C_Y(0) = \text{Var}(X_j) + \frac{1}{4}\text{Var}(X_{j-1}) = 8000 + \frac{1}{4}(8000) = 10000.$$

For $|m| = 1$,

$$C_Y(m) = -\frac{1}{2}\text{Var}(X) = -\frac{1}{2}(8000) = -4000.$$

For $|m| > 1$,

$$C_Y(m) = 0.$$

Hence

$$C_Y(m) = \begin{cases} 10000, & m = 0, \\ -4000, & |m| = 1, \\ 0, & |m| > 1. \end{cases}$$

Autocorrelation

Using $R_Y(m) = E[Y_j Y_{j+m}] = C_Y(m) + \mu_Y^2$ and $\mu_Y = 200$,

$$R_Y(m) = \begin{cases} 50000, & m = 0, \\ 36000, & |m| = 1, \\ 40000, & |m| > 1. \end{cases}$$

(4.2) X_i Gaussian WSS, $E[X_i] = 400$,

$$C_X(m) = 8000e^{-|m|}$$

Mean

$$E[Y_i] = 400 - \frac{1}{2}(400) = 200.$$

Therefore,

$$\mu_Y = 200.$$

Autocovariance

For lag m ,

$$\begin{aligned} C_Y(m) &= \text{Cov}(Y_i, Y_{i+m}) \\ &= \text{Cov}\left(X_i - \frac{1}{2}X_{i-1}, X_{i+m} - \frac{1}{2}X_{i+m-1}\right) \\ &= C_X(m) - \frac{1}{2}C_X(m-1) - \frac{1}{2}C_X(m+1) + \frac{1}{4}C_X(m). \end{aligned}$$

Thus

$$C_Y(m) = \frac{5}{4}C_X(m) - \frac{1}{2}C_X(m-1) - \frac{1}{2}C_X(m+1).$$

Substituting $C_X(m) = 8000e^{-|m|}$,

$$C_Y(m) = 10000e^{-|m|} - 4000e^{-|m-1|} - 4000e^{-|m+1|}.$$

In particular, the variance is

$$\text{Var}(Y_i) = C_Y(0) = 10000 - 8000e^{-1}.$$

So

$$\text{Var}(Y_i) = 10000 - 8000e^{-1}.$$

Autocorrelation

Again,

$$R_Y(m) = C_Y(m) + \mu_Y^2 = C_Y(m) + 40000.$$

Hence

$$R_Y(m) = 40000 + 10000e^{-|m|} - 4000e^{-|m-1|} - 4000e^{-|m+1|}.$$

(4.3) Joint pdf of Y_2 and Y_3

Because (X_i) is a Gaussian process and (Y_2, Y_3) is a linear transformation of Gaussian variables, (Y_2, Y_3) is jointly Gaussian.

Its mean vector is

$$\boldsymbol{\mu} = \begin{bmatrix} 200 \\ 200 \end{bmatrix}.$$

Its covariance matrix is

$$\Sigma = \begin{bmatrix} \sigma_Y^2 & C_Y(1) \\ C_Y(1) & \sigma_Y^2 \end{bmatrix},$$

where

$$\sigma_Y^2 = C_Y(0) = 10000 - 8000e^{-1},$$

and

$$C_Y(1) = 10000e^{-1} - 4000 - 4000e^{-2}.$$

Therefore, the joint pdf is

$$f_{Y_2, Y_3}(y_2, y_3) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp \left(-\frac{1}{2} [y_2 - 200 \quad y_3 - 200] \Sigma^{-1} \begin{bmatrix} y_2 - 200 \\ y_3 - 200 \end{bmatrix} \right).$$

Question 5

Since X and Y are jointly Gaussian, any linear combination is also Gaussian. Let

$$Z = X - 2Y.$$

Then

$$E[Z] = E[X] - 2E[Y] = 5 - 2(2) = 1.$$

Also,

$$\text{Var}(Z) = \text{Var}(X) + 4 \text{Var}(Y) - 4 \text{Cov}(X, Y).$$

(5.1) X and Y uncorrelated

Since jointly Gaussian and uncorrelated implies independent,

$$\text{Cov}(X, Y) = 0.$$

Hence

$$\text{Var}(Z) = 0.2 + 4(0.4) = 1.8.$$

So

$$Z \sim \mathcal{N}(1, 1.8).$$

Therefore,

$$f_Z(z) = \frac{1}{\sqrt{2\pi(1.8)}} \exp \left(-\frac{(z-1)^2}{2(1.8)} \right), \quad -\infty < z < \infty.$$

(5.2) Correlation coefficient $\rho = 0.4$

Now

$$\text{Cov}(X, Y) = \rho \sigma_X \sigma_Y = 0.4\sqrt{0.2}\sqrt{0.4}.$$

Thus

$$\text{Var}(Z) = 0.2 + 4(0.4) - 4 \left(0.4\sqrt{0.2}\sqrt{0.4} \right).$$

So

$$Z \sim \mathcal{N} \left(1, 0.2 + 1.6 - 1.6\sqrt{0.08} \right).$$

Equivalently,

$$\text{Var}(Z) = \frac{45 - 8\sqrt{2}}{25}.$$

Therefore,

$$f_Z(z) = \frac{1}{\sqrt{2\pi \left(\frac{45-8\sqrt{2}}{25} \right)}} \exp \left(-\frac{(z-1)^2}{2 \left(\frac{45-8\sqrt{2}}{25} \right)} \right), \quad -\infty < z < \infty.$$